



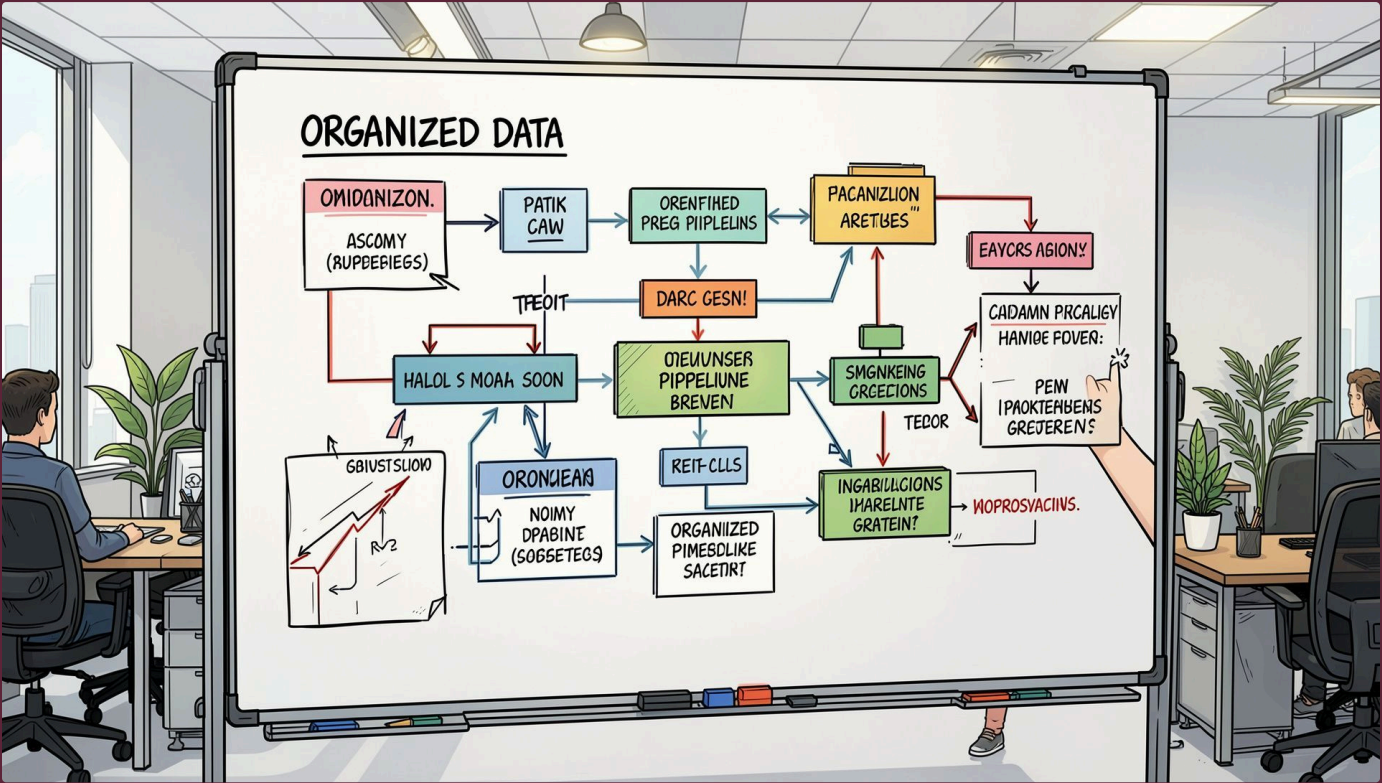
Customer Shopping Behavior Analysis

End-to-End Data Pipeline: Python ETL → PostgreSQL → SQL Analysis → Power BI Dashboard

DATA ANALYTICS

PYTHON · SQL · POWER BI

Project Overview



This project builds a complete analytics pipeline covering the full lifecycle of a real-world workflow: extracting and cleaning raw transactional data with Python, loading it into a PostgreSQL database, performing EDA and business queries in SQL, and connecting Power BI to build a live interactive dashboard.

3,900 records

Customer purchases

18 columns

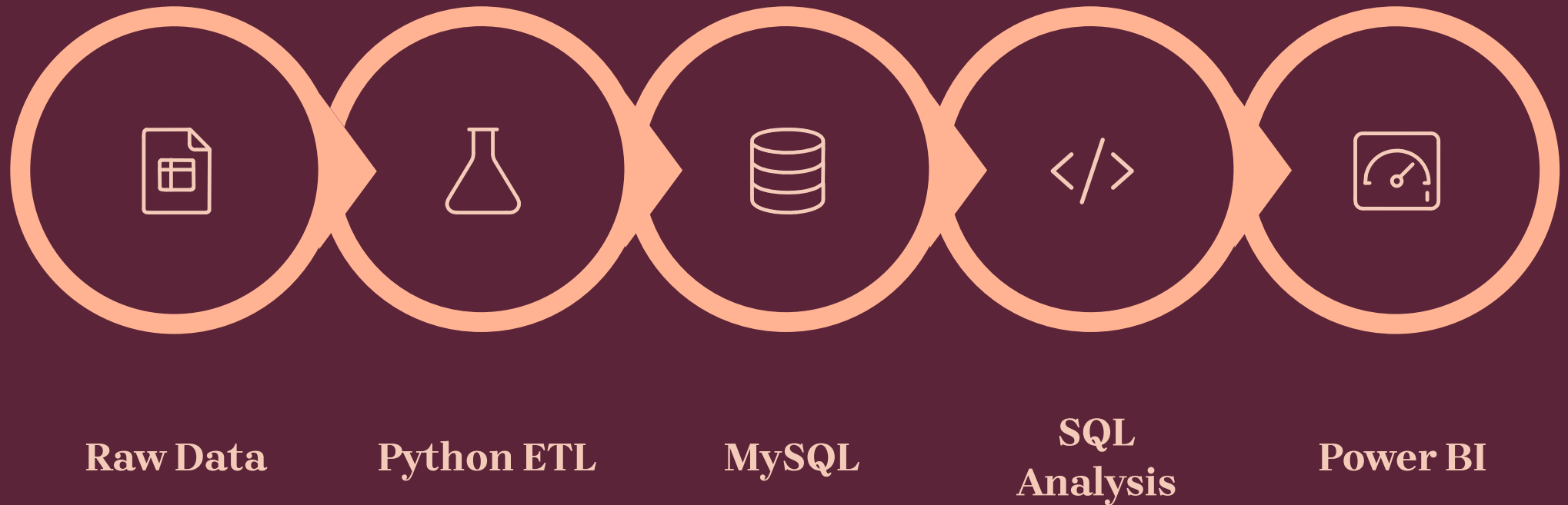
Demographics & behavior

Tech Stack

Layer	Tool	Purpose
Extraction & Transformation	Python (pandas)	Clean, impute, feature engineer
Database	MySQL	Central source of truth
DB Connectivity	SQLAlchemy	Connect Python to MySQL
Analysis	SQL (PostgreSQL)	EDA & business queries
Visualization	Power BI Desktop	Live interactive dashboard

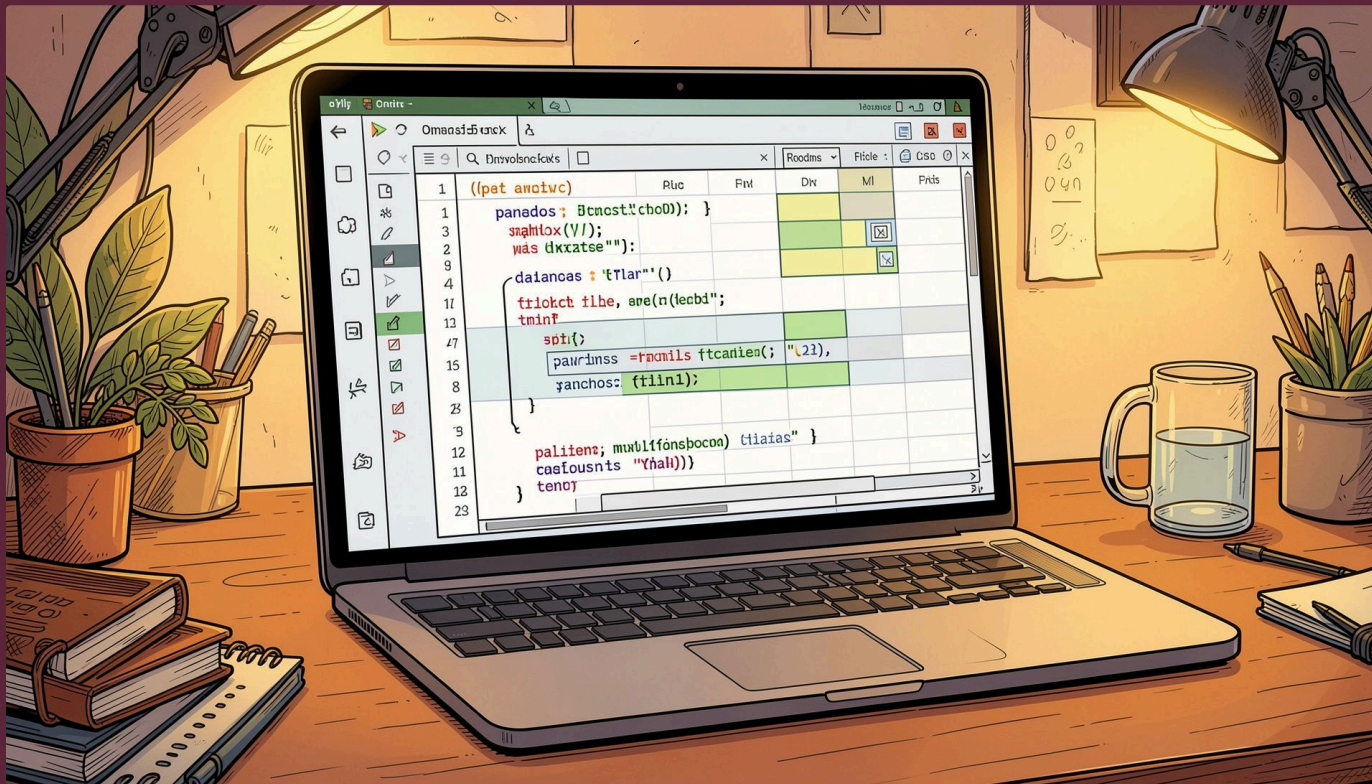
Pipeline Architecture

A five-stage linear flow with MySQL at the center as the single source of truth for both SQL analysis and Power BI.



Each stage feeds the next, ensuring data quality and consistency from raw CSV through to a live, interactive dashboard — with no manual re-exports required.

Phase 1 — ETL Using Python



→ Data Loading

Imported dataset with pandas; used `df.info()` and `.describe()` for initial exploration.

→ Missing Data

Imputed 37 missing Review Rating values using the median rating per product category.

→ Standardization

Renamed columns to `snake_case`; dropped redundant `promo_code_used` column.

→ Feature Engineering

Created `age_group` by binning ages; derived `purchase_frequency_days` from purchase data.

Phase 2 — Loading into MySQL

The cleaned DataFrame was loaded into MySQL so all subsequent analysis runs as SQL against a real database — not in-memory pandas. This also enables Power BI to connect to the same source of truth.

1

Connect

Connected to MySQL from Python using SQLAlchemy.

2

Create Schema

Built a table matching the cleaned schema (18 columns including `age_group`).

3

Load Data

Used `df.to_sql()` to load the cleaned DataFrame — eliminating the need to re-run Python cleaning for every analysis.

Phase 3 — SQL Analysis: Key Findings

Ten business-transaction queries run directly in MySQL.

\$157K

Male Revenue

vs. \$75K female — more than double

3,116

Loyal Customers

Vast majority of the base; retention over acquisition

~49%

Discount Dependency

Hat, Sneakers, Coat — margin risk

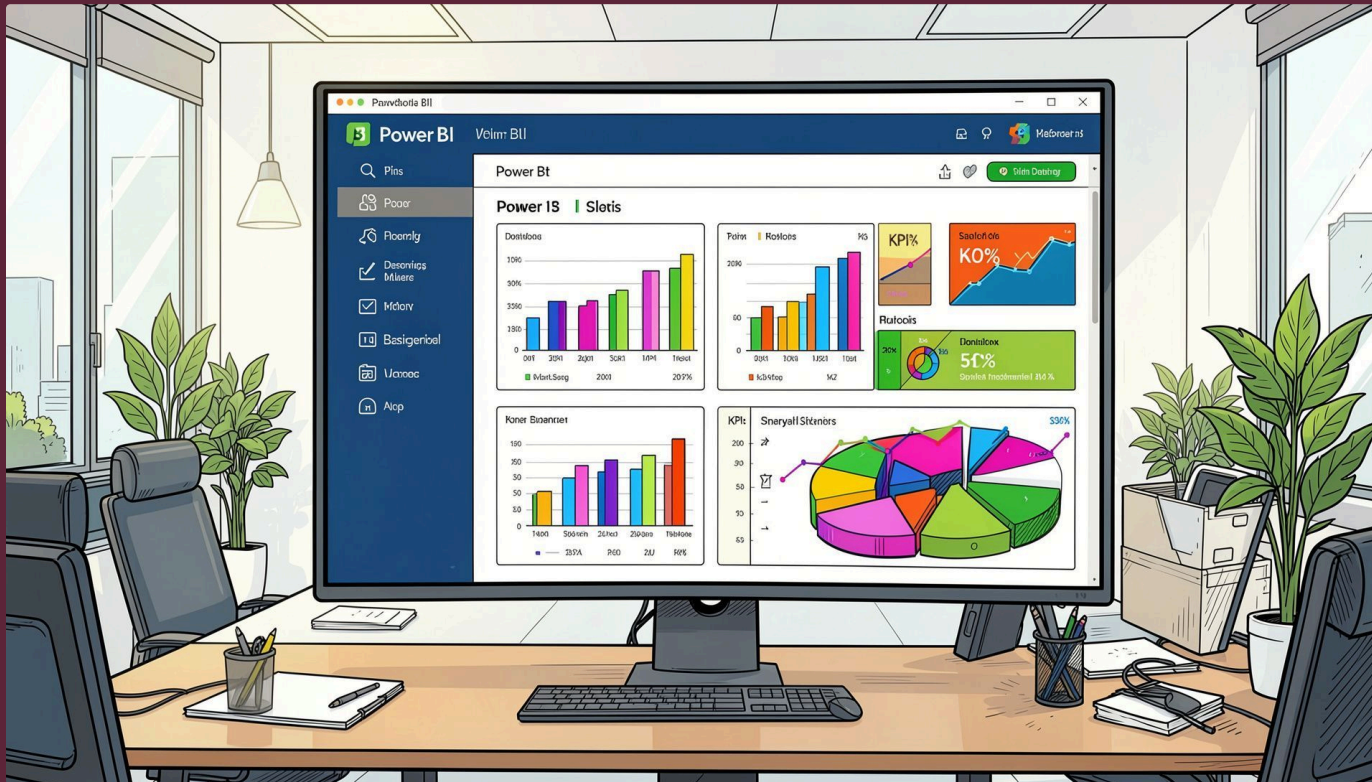
26.88%

Subscribers

Yet avg spend is flat vs. non-subscribers

❏ Revenue is evenly spread across age groups — age is not a strong differentiator on its own.

Phase 4 — Power BI Dashboard



Power BI connected directly to MySQL via its native SQL connector — no static CSV imports. The dashboard stays synced with the same data used for SQL analysis.

- **KPI Cards:** 4K customers, \$233K revenue, 3.75 avg rating, \$60 avg purchase
- **Slicers:** Subscription Status, Gender, Category, Shipping Type
- **DAX Measures:** Built for all key metrics
- **Charts:** Monthly revenue trend, category performance, payment methods, top cities, item colors, clothing sizes

Key Insights & Business Recommendations

Review Discount Policy

Hat, Sneakers, Coat are ~49–50% discount-dependent — test demand elasticity before cutting.

Focus on Retention

3,116 Loyal customers — invest in keeping them rather than early-funnel acquisition.

Targeted Marketing

Prioritize male customers and express-shipping users; they show the strongest revenue signals.

Re-examine Subscriptions

Avg spend is flat between subscribers and non-subscribers — the program isn't proven to drive higher spend.



Conclusion

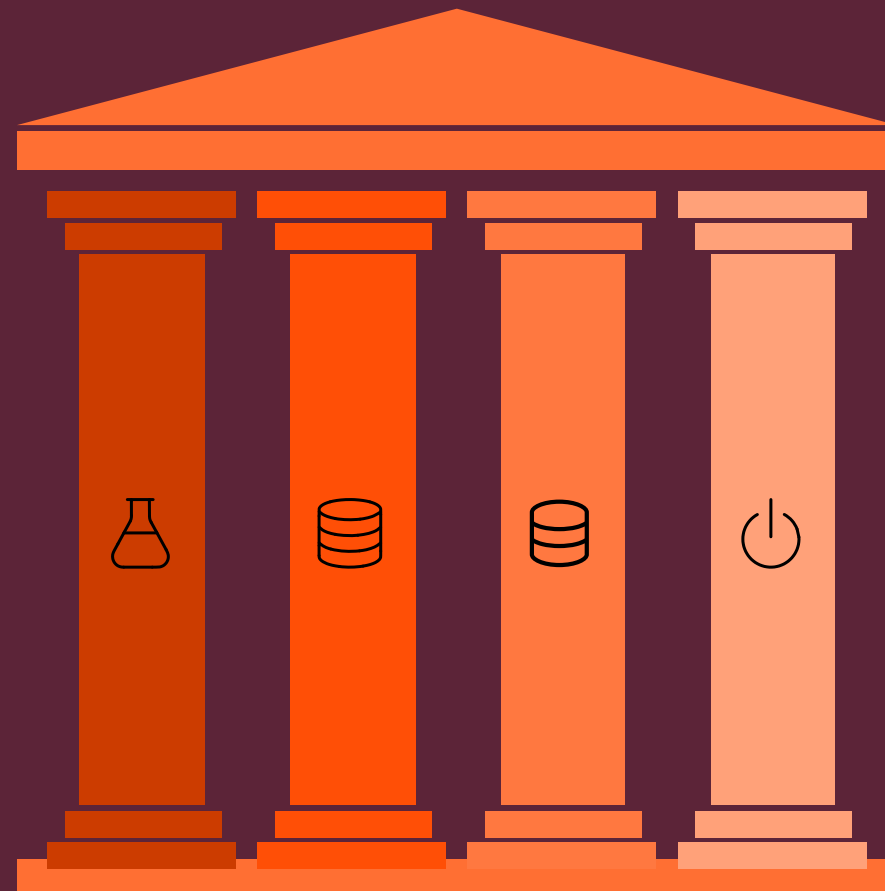
This project demonstrates a complete analytics pipeline from raw data to a live business dashboard. Because Power BI reads directly from MySQL rather than a static file, any future data update automatically flows through to the dashboard — no manual re-export required.

Python ETL

Cleanse and validate raw data

SQL Analysis

Structured business queries and metrics



MySQL Source

Central queryable source of truth

Power BI Live

Real-time dashboards connected to MySQL

✓ A reproducible, end-to-end workflow that scales — from messy CSV to actionable business intelligence.